The Seahorse Lookalike That Is Not a Seahorse at All

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At first glance, this thumb-sized fish looks exactly like a seahorse, with a horse-shaped head tilted downward, a curled grasping tail, and, in males, a fully enclosed pouch for the young. But it is not a seahorse, scientists report in a new study in the Journal of Lost Species. They describe Australyichthys **aegis**, a fish from deep reefs off southern Australia, and call it the helmethead sea muppet for the domed, helmet-like crest on its head. In Greek mythology, the **aegis** was a protective shield. With two relatives, the bullneck sea muppet and the paradoxical sea muppet, it forms a small group nicknamed the sea muppets. All three had been placed in the seahorse **genus** for their looks, but scientists now place them on their own branch of the family tree, in a new **genus**, Australyichthys, from Latin for southern and Greek for fish. The whole **genus** is known from just seven specimens gathered over about 65 years across more than 2,500 kilometers of coastline, and nothing has been collected in nearly two decades. You will not find them scuba diving: each was brought up by accident in trawl nets or dredges between about 50 and 125 meters down. The A. **aegis** specimen came up in 2006 near the Gambier Islands and was mistakenly filed in an **invertebrate** collection for years. Because the specimens are irreplaceable, scientists could not dissect them. Instead, **micro-CT** scans built three-dimensional models of their skeletons for bone-by-bone comparison with true seahorses. True seahorses have a rigid cage of fused belly plates and a bony crown called a **coronet**; Australyichthys has neither. DNA from the one ethanol-preserved specimen placed it outside seahorses, on their closest relative branch. Biologists call this look-alike pattern **convergent evolution**.

Word Treasure

aegis -- In Greek mythology, the aegis was a protective shield.

genus -- A genus is a small group of closely related living things.

coronet -- A coronet is a bony crown on top of a seahorse's head.

convergent evolution -- Convergent evolution is a pattern where look-alike animals develop similar features.

micro-CT -- Micro-CT is a kind of scan that builds a three-dimensional model of a skeleton.

invertebrate -- An invertebrate is an animal without a backbone.

Background you need

A genus is a group of closely related species, and a family tree in biology shows how different groups are related to each other.

Animals can look very similar even when they are not close relatives, because they may have developed the same useful features separately, a pattern called convergent evolution.

True seahorses are set apart by special body features, such as a rigid cage of fused belly plates and a bony crown on the head.

Scientists compare skeletons and DNA to decide where a species belongs, even when its outside appearance points in a different direction.

Break it down

LEAD: A thumb-sized fish that looks exactly like a seahorse is not a seahorse at all.

KEY EVENT: Scientists describe Australyichthys aegis, the helmethead sea muppet, in a new study in the Journal of Lost Species.

+- WHO: Australyichthys aegis and its two relatives, the bullneck sea muppet and the paradoxical sea muppet

+- WHERE: Deep reefs off southern Australia, brought up from between about 50 and 125 meters down

+- WHEN: The A. aegis specimen came up in 2006 near the Gambier Islands

EVIDENCE: Micro-CT scans built three-dimensional models of their skeletons for bone-by-bone comparison with true seahorses, and DNA from the one ethanol-preserved specimen placed it outside seahorses, on their closest relative branch.

+- SEAHORSE TRAIT MISSING: Australyichthys has neither the rigid cage of fused belly plates nor the bony crown called a coronet found in true seahorses

+- PATTERN: Biologists call this look-alike pattern convergent evolution

CONTEXT: The whole genus is known from just seven specimens gathered over about 65 years across more than 2,500 kilometers of coastline, and nothing has been collected in nearly two decades.

Why it matters

It shows that animals can evolve similar shapes independently, so appearance alone cannot settle how species are related. Studying rare deep-reef fish like this helps scientists place species on the family tree more accurately.

Test what you remember

1. What common name did scientists give Australyichthys aegis?

- (A) The bullneck sea muppet
- (B) The southern seahorse
- (C) The helmethead sea muppet
- (D) The paradoxical sea muppet

2. Why is Australyichthys aegis called by a name linked to a protective shield?

- (A) Because it lives on deep reefs
- (B) Because it was found near the Gambier Islands
- (C) Because its tail can grasp objects
- (D) Because in Greek mythology, the aegis was a protective shield

3. How many specimens is the whole genus Australyichthys known from?

- (A) Seven
- (B) More than 2,500
- (C) Three
- (D) Sixty-five

4. How were the specimens collected?

- (A) From an invertebrate collection in 2006
- (B) From shallow coastal waters
- (C) By scuba divers
- (D) By accident in trawl nets or dredges

5. What did scientists use instead of dissecting the irreplaceable specimens?

- (A) Micro-CT scans that built three-dimensional models of their skeletons
- (B) Photographs of live fish
- (C) DNA from every specimen
- (D) A new trawl net

6. What term do biologists use for the look-alike pattern shown by Australyichthys and true seahorses?

- (A) A coronet
- (B) Convergent evolution
- (C) The seahorse genus
- (D) A fused belly plate

Answer key (try the quiz first!): 1: C 2: D 3: A 4: D 5: A 6: B

Questions to think about

1. Scientists studying the fish: They place the sea muppets on their own branch of the family tree, in a new genus, Australyichthys, rather than in the seahorse genus.

MY THOUGHTS (at least 20 words):